

Nisal V. Kalpage

ELECTRICAL ENGINEER · RESEARCHER · PHD CANDIDATE

Department of Electrical and Computer Systems Engineering,
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Profile

ELECTRICAL ENGINEER - ELECTRONIC ENGINEER - TELECOMMUNICATIONS ENGINEER - WIRELESS COMMUNICATION - SATELLITE COMMUNICATIONS - INTEGRATED SENSING AND COMMUNICATIONS - COMMUNICATION THEORY - SIGNAL PROCESSING - INFORMATION THEORY - NETWORK ENGINEERING

I am a motivated and detail-oriented Communications Engineer with a strong foundation in electrical and telecommunications engineering, complemented by excellent interpersonal skills and a structured approach to documentation and project coordination. My background combines advanced technical training in communications systems, wireless technologies, and network theory with practical exposure to real-world infrastructure environments.

Through my academic and industry experience, I have contributed to system design, performance analysis, technical documentation, and multi-disciplinary project coordination. I have developed strong capabilities in communications architecture, WiFi modelling, structured systems design, and analytical problem-solving. I work effectively within team environments, interpret technical requirements clearly, and deliver organised, high-quality outputs aligned with project objectives. I am particularly motivated to grow within a consulting environment where technical expertise is combined with client engagement and practical design delivery.

Education

Monash University

Clayton, Victoria, Australia

PHD CANDIDATE, DEPARTMENT OF ELECTRICAL AND COMPUTER SYSTEMS ENGINEERING

July 2022 – Present

- Research Topic: "Orthogonal Time Frequency Space Modulation"
- Advisor: A/Prof. Yi Hong
- Collaborators: Dr. Preety Priya, Dr. Viduranga Wijekoon

Curtin University and SLIIT (2+2)

Colombo, Sri Lanka

B.ENG. IN ELECTRICAL AND ELECTRONIC ENGINEERING (HONOURS)

Feb. 2018 – Nov. 2021

- Course Weighted Average (CWA): 87.97
- Class: First Class Honours (Top of the class)
- Thesis: Orthogonal Frequency Division Multiplexing Transceiver Design with SystemC and an FPGA Implementation
- Advisor: Dr. Rohana Thilakumara

Experience

Teaching Associate/ Associate Lecturer

Clayton, Victoria, Australia

MONASH UNIVERSITY

Feb. 2023 – Present

- ECE5883/ 6883 - Advanced Signal Processing (Deputy Unit Coordinator)
- ECE5884/6884 - Wireless Communications (Deputy Unit Coordinator)
- ECE3142 - Digital Communications (Deputy Unit Coordinator)
- ECE3161 - Analog Electronics
- ECE4042 - Communications Theory
- ECE3141 - Information and Networks
- FIT2100 - Operating Systems
- ECSE Help Desk

University Associate / Academic Instructor

Colombo, Sri Lanka

CURTIN UNIVERSITY - SLIIT INTERNATIONAL

Feb. 2022 – July 2022

- Introduction to MATLAB - Laboratory Instructor
- Mechanics Lab - Laboratory Instructor
- Electrical Systems

Intern - Network Operations and Service Assurance

Colombo, Sri Lanka

DIALOG AXIATA PLC

Dec. 2020 – Mar. 2021

- Introduction to Data Centre Operation and Management
- Introduction to Data Centre Security Systems
- Introduction to Data Centre Power Management and Redundancy Mechanisms

- Engaged in the development of the OREL Smart Home IoT project
- Development and testing of a Power Guard PCB Electronic Design
- Involved in general Research and Development activities

Honours & Awards

Faculty of Engineering Post Publication Award , <i>Faculty of Engineering, Monash University</i>	July. 2026
Competitive award awarded to PhD students after submitting their thesis.	
Peoples Choice Award 3MT - Department Level , <i>ECSE, Monash University</i>	May. 2025
Awarded the peoples choice the department level 3 Minute Thesis competition.	
Best Presentation , <i>Monash University</i>	Nov. 2024
Award for best presentation in the Software and Hardware category.	
2nd Place 3MT - Faculty Level , <i>Faculty of Engineering, Monash University</i>	Jul. 2024
Awarded the 2nd place for the faculty level 3 Minute Thesis competition after a very close competition.	
1st Place 3MT - Department Level , <i>ECSE, Monash University</i>	May. 2024
Awarded the 1st place for the department level 3 Minute Thesis competition.	
Monash Graduate Scholarship , <i>Monash University</i>	Jul. 2022
Monash Graduate Scholarships are granted to students pursuing a research doctorate in recognition of their outstanding academic achievements.	
Monash International Tuition Scholarship , <i>Monash University</i>	Jul. 2022
Awarded to international students pursuing a research doctorate, based on their academic record, research outputs, and prior research experience.	
Best All Rounder , <i>Curtin Alumni Sri Lanka Chapter</i>	Mar. 2022
Awarded the best all rounder award for exemplary academic and extra curricular activities during undergraduate studies in the Faculty of Engineering.	
Curtin Award of Excellence (Electrical and Electronic Engineering) , <i>Curtin University (Sri Lanka)</i>	Mar. 2022
Awarded for the best academic performance in the Faculty of Engineering.	
Valedictorian Speaker , <i>Curtin University (Sri Lanka) Convocation</i>	Mar. 2022
Honoured in giving the valedictorian speech for the graduating class of 2022.	
Curtin Extra Award , <i>Curtin University</i>	Nov. 2021
Awarded for representing students of our Faculty.	
Vice Chancellors List Award , <i>Curtin University</i>	Jun. 2020
Awarded for exemplary academic performance being in the top 1% of undergraduates in all Curtin University campuses.	
Curtin University Letter of Commendation , <i>Curtin University</i>	2020, 2021
Awarded for weighted Average Mean (WAM) greater than 80% for Semester 1 and 2 of 2020 and Semester 1 and 2 of 2021	
Deans List Award , <i>Sri Lanka Institute of Information and Technology</i>	2018, 2019
Awarded for GPA greater than 3.7/4 for Semester 1 of 2018 and Semester 1 and 2 of 2019	
SLIIT Scholarship , <i>Sri Lanka Institute of Information and Technology</i>	Jul, 2018, 2019
Awarded for academic performance being the best student in the Faculty of Engineering	

Projects

Orthogonal Frequency Division Multiplexing Transceiver Design with SystemC and an FPGA Implementation

Development of an OFDM Transceiver model through SystemC and Simulation through VHDL development, development of MATLAB models for testing and validation of system design.

LANGUAGES, TOOLS AND SOFTWARES: *SystemC, VHDL, Verilog, FPGA, C++, C, MATLAB*

Development of a EEPROM Memory Levelling Algorithm

Development of a memory levelling algorithm to increase the number of read-write cycles possible for the EEPROM of a NodeMCU as a part of the Smart Light IoT project as a trainee engineer.

LANGUAGES, TOOLS AND SOFTWARES: *Arduino, C++, NodeMCU*

OREL Smart Home IoT Project

Engaged in the development of the OREL Smart Home IoT project and contributed in the development of the Smart Bulb, Smart Roller Door and Smart Roller Door IoT project components as a trainee engineer. Developed and tested embedded systems for functionality, reliability and

performance.

LANGUAGES, TOOLS AND SOFTWARES: [Arduino](#), [C++](#), [json](#), [OTA Updates](#), [Google IoT](#), [MQTT](#)

OREL Power Guard Design Project

Research on the operation and functionality of Power Guard Device and developing, implementing and testing a Power Guard device.

LANGUAGES, TOOLS AND SOFTWARES: [Altium](#), [PCB](#), [Viac](#)

RF Based Smart Lock Design and Development

Design and development of a key-less entry system for household use by using Radio Frequency transmissions and embedded system design.

LANGUAGES, TOOLS AND SOFTWARES: [Arduino](#), [Altium](#), [PCB](#), [RF](#)

LabView and MATLAB-Simulink based USRP testing

Implementation, evaluation and testing of different waveforms, modulation techniques, receiver implementations and other RF implementations on software based radio.

LANGUAGES, TOOLS AND SOFTWARES: [LabView](#), [MATLAB](#), [Simulink](#), [SDR](#), [USRP](#)

Publications

Noise Whitening and Error Performance Analysis of Oversampled DCT based OTFS

Under preparation

NISAL V. KALPAGE, PREETY PRIYA AND YI HONG

Performance Analysis of Imperfect Delay and/or Doppler Shifts for OTFS

*Under consideration for IEEE
Transactions on Vehicular
Technology*

NISAL V. KALPAGE AND YI HONG

Performance Analysis and Joint Receiver Design for an OTFS-Based ISAC System in Time-Varying Channels

*IEEE Transactions on Vehicular
Technology*

NISAL V. KALPAGE, YI HONG AND PREETY PRIYA

DCT-Based OTFS With Reduced PAPR

IEEE Communication Letters

NISAL V. KALPAGE, PREETY PRIYA AND YI HONG

Seminars and Talks

Error Performance Analysis of OTFS with Imperfect Channel Estimation

Sep. 2025

Department Electrical and Computer Systems Engineering, Monash University

PAPR ANALYSIS OF DCT BASED OTFS

Nov. 2024

Monash Engineering Postgraduate Conference, Monash University

Orthogonal Time Frequency Space Fundamentals and Applications

Sep. 2024

Department Electrical and Computer Systems Engineering, Monash University

Performance Analysis and Joint Receiver Design for an OTFS-Based ISAC System in Time-Varying Channels

Jul. 2024

Department Electrical and Computer Systems Engineering, Monash University

DCT-Based OTFS With Reduced PAPR

Jul. 2023

Department Electrical and Computer Systems Engineering, Monash University

Services

Journal Peer Reviewer

- IEEE Communication Letters
- IEEE Transactions on Communications
- IEEE Transaction on Signal Processing
- IEEE Transaction on Wireless Communications

Final Year Project Supervision

- Waveform Design for Wireless Communication
- OTFS Transmitter Performance Analysis

Extra Curricular Activities

- **President** - ECSE PG Community Monash University
- **Organising Committee Member** - Monash University 4th Engineering Postgraduate Conference
- **Executive Committee Member** - Curtin University Alumni Sri Lanka Chapter
- **President** - Faculty of Engineering Student Community (FESC)
- **Vice-President** - Engineering Research and Innovations Club (ERIC)
- **Treasurer** - Institute of Electrical and Electronics Engineers Student Branch SLIIT (IEEE)
- **Member** - Curtin International Student Leaders (CISL) Working Group
- **Member of Finance Team** - Techno Meetup Sri Lanka, IEEE Sri Lanka Section